

Wall256 — SU(3) Strong-Coupling Conditional Reduction

An honest report · what is machine-checked, what stays open, and what is NOT proven

CONDITIONAL REDUCTION — NO CLAY RESULT IS PROVEN

This document describes a pure logical REDUCTION. It does NOT prove a Yang-Mills mass gap, a spectral gap, convergence of any real cluster expansion, or close Clay Surface #1. The entire mathematical content lives in THREE explicit open hypotheses; none is discharged. Scope is the LATTICE SU(3) strong-coupling regime, NOT the Clay continuum problem.

What this is

Towers/YM/Wall256_Scaffold.lean is a single sorry-free, axiom-clean theorem (strong_coupling_decay_of_open_inputs). It threads the three open inputs of the Osterwalder-Seiler strong-coupling analysis through already-landed, genuinely machine-checked algebra (comparison-test summability + a $\rho^d = \exp(-\Delta \cdot d)$ identity) to an abstract two-point decay shape. The conclusion holds ONLY IF all three hypotheses hold.

PROVEN

The reduction wiring

comparison-test summability + $\rho^d = \exp(-\Delta \cdot d)$ algebra are machine-checked.

OPEN

The 3 hypotheses

SU(3) Haar bound, Osterwalder-Seiler cluster step, Brydges-Federbush bridge.

NONE

Mass-gap claim

No $\mu > 0$, no spectral gap, no Surface #1, no continuum / Clay statement.

The exact Lean statement

Verbatim from Wall256_Scaffold.lean — the only mathematical assertion in this file

```
theorem strong_coupling_decay_of_open_inputs
  {E : Type*} (corr sep : E → E → ℝ) (C ρ w1 : ℝ)
  {N a : ℕ → ℝ} (hN0 : ∀ n, 0 ≤ N n) (hN : ∀ n, N n ≤ (7 : ℝ) ^ n)
  (hw1 : w1 < 1 / 7)
  (h0S : w1 < 1 / 7 → TruncatedActivityBound a)
  (h_bridge : Summable (fun n : ℕ => N n * a n) →
    0 < ρ ∧ ρ < 1 ∧ ∀ x y, |corr x y| ≤ C * ρ ^ (sep x y)) :
  ∃ Δ : ℝ, 0 < Δ ∧ ∀ x y, |corr x y| ≤ C * Real.exp (-Δ * sep x y) :=
  su2_gap_of_truncatedActivity corr sep C ρ hN0 hN (h0S hw1) h_bridge
```

[OPEN](#)[OPEN](#)[OPEN](#)

In plain language

- `corr`, `sep` are ABSTRACT — an arbitrary correlator and separation on any type `E`. No real Wilson-loop correlator or lattice metric is built.
- `hN0`, `hN`: the polymer entropy count `N n` is bounded by the 7^n geometric growth (the $7 = \text{SU}(3)$ coordination of connected polymers through a fixed plaquette).
- Conclusion: exponential two-point decay $|\text{corr}| \leq C \cdot \exp(-\Delta \cdot \text{sep})$ with $\Delta > 0$ — IF `H1`, `H2`, `H3` all hold.
- The proof body is one line: it just plugs the three hypotheses into the already-landed reduction combinator `su2_gap_of_truncatedActivity` (named for legacy reasons; it is group-agnostic and asserts no gap).

The three OPEN inputs

Each is a hypothesis — never proved here. The real-world theorem each stands for is named.

H1 SU(3) single-site Haar weight (strict bound)

OPEN

`hw1 : w1 < 1/7`

`w1` stands for $\int_{\{SU(3)\}} \exp(-\beta \cdot \text{actL}) d(\text{haar})$. The strict bound $< 1/7$ holds for large β ($\beta > \sim 0.85$), a genuine Haar / character-expansion estimate. `mathlib v4.12.0` cannot evaluate it, so it is carried as a real-number hypothesis on an abstract `w1`.

Real-world source: SU(3) character theory / verified cubature (absent from `mathlib v4.12.0`).

H2 Osterwalder-Seiler 1978, Thm 2.1 (Ursell / cluster step)

OPEN

`h0S : w1 < 1/7 → TruncatedActivityBound a`

Single-site smallness propagates, via the truncated (Ursell) cluster expansion, to a per-size connected-polymer activity bound $a_n \leq \exp(-l)^n$ with rate $l > \log 7$. The cluster expansion is absent from `mathlib`, so this implication is a hypothesis.

Real-world source: Osterwalder & Seiler, *Ann. Phys.* 110 (1978); truncated polymer expansion.

H3 Brydges-Federbush KP bridge

OPEN

`h_bridge : Summable(∑ N n · a n) → (0 < ρ < 1 ∧ clustering)`

Kotecký-Preiss summability of the entropy-weighted polymer series turns into geometric two-point clustering with spectral radius $\rho < 1$. Standard textbook cluster-expansion theory, but absent from `mathlib v4.12.0`; carried as a hypothesis (NOT by sorry).

Real-world source: Friedli & Velenik, *Statistical Mechanics of Lattice Systems* (2018), Ch. 5.

Why the bound must be STRICT: $w_1 < 1/7$, not $w_1 = 1/7$

The boundary case makes the polymer entropy series diverge — the reduction would be vacuous

The entropy vs. smallness race

Connected polymers of size n through a fixed plaquette number at most 7^n (entropy). Each carries an activity suppressed at rate $\exp(-I)^n$. Convergence of $\sum 7^n \cdot \exp(-I)^n$ needs $\exp(-I) < 1/7$, i.e. $I > \log 7$.

$$w_1 < 1/7$$

$$\Rightarrow I > \log 7$$

$\exp(-I) < 1/7$, so the geometric ratio $7 \cdot \exp(-I) < 1$.

$$\sum 7^n \cdot \exp(-I)^n$$

$$= \sum (7 \cdot \exp(-I))^n$$

converges (geometric, ratio < 1) — KP summable.

$$w_1 = 1/7$$

$$\Rightarrow I = \log 7$$

$\exp(-I) = 1/7$, ratio $= 7 \cdot (1/7) = 1$.

$$\sum 7^n \cdot (1/7)^n$$

$$= \sum 1$$

DIVERGES — the comparison test fails; the reduction is vacuous.

Consequence: the boundary $\beta = 0.85$ (where $w_1 = 1/7$) is EXCLUDED. Only the strict interior bound gives a convergent polymer series. This is exactly the gap between necessary and sufficient — and it is precisely the content that mathlib cannot supply.

Wall256_Scaffold — conditional reduction tower

strong_coupling_decay_of_open_inputs · SU(3) lattice strong-coupling · proves NO gap

H1

OPEN

hw1 : w1 < 1/7

SU(3) single-site Haar weight
strict bound.

$w1 = \int \exp(-\beta \cdot \text{actL}) d \text{ haar}.$
 $= 1/7 \Rightarrow I = \log 7$, then
 $\sum_n 7^n (1/7)^n$ diverges; $\beta=0.85$ excl.

H2

OPEN

h0S : w1 < 1/7 $\rightarrow$ TruncatedActivityBound a

Osterwalder-Seiler (1978, Thm 2.1)
Ursell / cluster step: single-site smallness
 $\Rightarrow$ polymer activity rate $I > \log 7$.

H3

OPEN

h_bridge

Brydges-Federbush. KP-summability
Summable $(\sum_n N n \cdot a n) \Rightarrow 0 < \rho \wedge \rho < 1$
 $\wedge$ geometric two-point clustering.

strong_coupling_decay_of_open_inputs

machine-checked threading · classical trio · SORRY = 0

PROVEN edges:

$\text{kp_summable_of_truncatedActivity}$
 $\rho^d = e^{(-\Delta \cdot d)} \text{ (mass_gap_pos_of_spectral_gap)}$

Conditional two-point decay

$\exists \Delta > 0, \forall x y, |\text{corr } x y| \leq C \cdot \exp(-\Delta \cdot \text{sep } x y)$

Holds ONLY IF all three OPEN inputs are discharged. LATTICE scope, NOT the Clay mass gap.

Wall256 — derivation in language (annotated)

symbols $\Rightarrow$ plain language · each arrow tagged PROVEN (machine-checked) or OPEN (hypothesis)

$$w1 < 1/7$$

single-site Haar weight is small

[H1] **OPEN**

$$\Rightarrow a_n \leq e^{(-I \cdot n)}, \text{ rate } I > \log 7$$

Osterwalder-Seiler: smallness propagates to a per-size polymer activity bound

[H2] **OPEN**

$$\Rightarrow \sum_n N_n \cdot a_n \leq \sum_n 7^n \cdot e^{(-I \cdot n)} = \sum_n (7 \cdot e^{(-I)})^n < \infty$$

term-by-term comparison test; converges since $I > \log 7 \Leftrightarrow 7 \cdot e^{(-I)} < 1$

↳ hinge — strict $< 1/7$ is essential: $= 1/7 \Rightarrow I = \log 7 \Rightarrow \sum_n 1$ diverges

kp_summable_of_truncatedActivity **PROVEN**

$$\Rightarrow 0 < \rho < 1 \wedge |\text{corr } x \ y| \leq C \cdot \rho^{(\text{sep } x \ y)}$$

Brydges-Federbush: summability $\Rightarrow$ geometric two-point clustering

[H3] **OPEN**

$$\Rightarrow \Delta = -\log \rho > 0, |\text{corr } x \ y| \leq C \cdot e^{(-\Delta \cdot \text{sep } x \ y)}$$

$\rho^d = e^{(-\Delta \cdot d)}$ algebra

mass_gap_pos_of_spectral_gap **PROVEN**

Holds ONLY IF H1-H3 are discharged. LATTICE scope, NOT the Clay mass gap. corr / sep are abstract.

$w1 < 1/7$ $I > \log 7 \approx 1.9459$ $\beta = 0.85$ boundary EXCLUDED entropy count $N_n \leq 7^n$

The parallel in the NS tower (for contrast)

The same honesty pattern governs Navier-Stokes — and NS is FROZEN at the Clay boundary

Same shape, different equation

Wall256 reduces a lattice gap to an open clustering bound. The NS tower analogously reduces global regularity to an open a-priori estimate. In both, the hard analytic input is named, not proved.

Beale-Kato-Majda criterion

OPEN

Global smoothness holds iff $\int_0^T \|\omega(t)\|_\infty dt < \infty$ (ω = vorticity). The time-integrated vorticity sup is the open quantity — controlling it is equivalent to the regularity problem itself.

Enstrophy / vorticity control

OPEN

enstrophy_bound_global_Surface — a global-in-time bound on the vorticity seminorm. A genuine open surface; necessary input to BKM, not supplied.

NS FREEZE (locked invariant)

FROZEN

Towers/NS/* is frozen at milestone NS-540-phase6-clay-boundary. Surface #1 (global regularity) and Surface #2 (modeled weak existence) stay OPEN. No 'NS solved' / 'regularity proven' claim is made.

No cross-tower bridge is claimed: the YM lattice reduction and the NS regularity reduction are SEPARATE. Neither implies the other; neither is solved.

What is NOT proven

Read this page before drawing any conclusion from this report

- ✗ NO Yang-Mills mass gap. No $m > 0$, no $\Delta > 0$ unconditionally — Δ exists only under H1+H2+H3.
- ✗ NO spectral gap for any real Wilson transfer operator. corr/sep are abstract symbols.
- ✗ NO closure of Clay Surface #1. It stays OPEN; YM tower Status: Open.
- ✗ NO continuum / Clay statement. Scope is the LATTICE SU(3) strong-coupling regime only.
- ✗ NO convergence of a real cluster expansion. The KP summability is conditional on H2's rate $l > \log 7$.
- ✗ NO discharge of any sorry or named open surface. This file adds none and removes none.
- ✗ NO link to the $\pi/10$ exceptional-prime sieve, to κ_{15} , or to any number-theoretic 'gap'.
- ✗ NO cross-tower bridge between the YM and NS reductions.

What IS true

- ✓ The file is sorry-free and axiom-clean (classical trio {propext, Classical.choice, Quot.sound}).
- ✓ The reduction wiring — comparison-test summability and the $\rho^d = \exp(-\Delta \cdot d)$ algebra — is machine-checked.
- ✓ The three hard inputs are stated explicitly and honestly, each pointing at its real-world theorem.

Verdict: a clean conditional REDUCTION. The mass gap remains OPEN and unproven.

Towers/YM/Wall256_Scaffold.lean · classical trio · SORRY: 0 · NOT a brick · YM Status: Open